Transformer MLP Transplants Depend on the Receiving State Supplementary Material

Muhammed Nehan
Ajman University
202410365@ajmanuni.ac.ae

A Related Work

Activation transplantation and causal interventions. Activation patching, interchange interventions, causal tracing, path patching, and circuit discovery intervene on hidden states to identify the components that causally drive a behavior [Vig et al., 2020, Geiger et al., 2021, 2025, Goldowsky-Dill et al., 2023, Conmy et al., 2023, Wu et al., 2023, Zhang and Nanda, 2024]. AtP* approximates exhaustive patching with gradients [Kramár et al., 2024]. Patchscopes, Future Lens, function vectors, relational-concept extraction, and concept transplantation move information or behavior across contexts [Ghandeharioun et al., 2024, Pal et al., 2023, Todd et al., 2024, Wang et al., 2024, Dong et al., 2025]. Transfer is not automatic: the outcome of activation swaps depends on the protocol, several mediators can interact, causal vectors need not be invariant across formats, intervened representations can diverge from the model's natural distribution, and role claims about components can fail to transfer [Garcia, 2026, Vaidyanathan et al., 2026, Opiełka et al., 2026, Grant et al., 2026, Quirke, 2026]. Our design differs from these lines of work by holding an example-specific natural MLP displacement exactly fixed while changing only the natural state that receives it, so that the receiver's causal contribution is isolated.

MLP updates as units of computation. Feed-forward layers have been described as key–value memories and as promoting concepts in vocabulary space, and editing their weights is a standard way to change stored associations [Geva et al., 2021, 2022, Meng et al., 2022]. These views treat an MLP output as carrying content. We do not assign content to an MLP update; we measure how its causal effect depends on where it is written.

State-dependent steering and local geometry. Activation steering modifies hidden representations to change behavior [Turner et al., 2023, Rimsky et al., 2024, Li et al., 2023, Zou et al., 2023, Singh et al., 2024]. Recent work makes state dependence explicit through context-conditioned steering fields, local linear control, and response-theoretic sensitivity [Li et al., 2026, Skifstad et al., 2026, Olivieri and Pérez Rodríguez, 2026]. The softmax output has a natural Fisher geometry tied to behavioral change [Martens, 2020, Park et al., 2026], and FishBack pulls this geometry back to intermediate transformer states through the downstream Jacobian in order to steer [Wang and Zhao, 2026]. We do not propose a new geometry or a steering method. We use the established local metric as a diagnostic for whether the same natural update has different full-dose effects at different receivers.

Reusable internal computation. Hidden APIs asks whether hidden interfaces support reusable causal computations across forked futures [Ma et al., 2026]. Function vectors and related representation-transfer studies test whether particular behaviors remain portable across prompts or formats [Todd et al., 2024, Opiełka et al., 2026, Quirke, 2026]. Our transformation analysis does not assign a semantic function to an MLP update. It defines the computation operationally by its

NewInML Workshop at the 40th Conference on Neural Information Processing Systems (NeurIPS 2026), Paris, France, 2026.

measured causal footprint and asks whether receiver-specific structure in that footprint generalizes to unseen natural displacements.

B Mathematical Details

The identities in this appendix are standard. We state them to make the diagnostic and the baselines precise, not as new results.

B.1 Local KL identity

Let $A(z) = \log \sum_y e^{z_y}$, so that $\nabla A(z) = p = \mathrm{softmax}(z)$ and $\nabla^2 A(z) = C_p = \mathrm{diag}(p) - pp^\top$. For logits z and a logit change Δ,

$$D_{\mathrm{KL}}\big(\mathrm{softmax}(z) \,\big\|\, \mathrm{softmax}(z+\Delta)\big) = A(z+\Delta) - A(z) - \langle p, \Delta\rangle = \tfrac{1}{2}\,\Delta^\top C_p\,\Delta + O(\|\Delta\|^3). \quad (5)$$

Adding $\epsilon\Delta u$ to the receiver state gives $\Delta(\epsilon) = G_\ell(r + \epsilon\Delta u) - G_\ell(r) = \epsilon\, JG_\ell(r)\Delta u + O(\epsilon^2)$. Because the first-order term of (5) vanishes, the $O(\epsilon^2)$ part of Δ enters the divergence only at order ϵ^3, and

$$D_{\mathrm{KL}}(p_r \,\|\, p_{r+\epsilon\Delta u}) = \epsilon^2\, Q_{\mathrm{recv}}(r, \Delta u) + O(\epsilon^3), \quad (6)$$

with Q_{recv} as in (3). This is the classical second-order expansion of KL in a curved exponential family [Martens, 2020] pulled back through the downstream map [Park et al., 2026, Wang and Zhao, 2026]. The experiments evaluate the score at $\epsilon = 1$, where this expansion is not guaranteed to be accurate; whether it still ranks receivers is an empirical question.

B.2 Variance form and the matched baseline

For any $v \in \mathbb{R}^V$, $v^\top C_p v = \sum_y p_y v_y^2 - \big(\sum_y p_y v_y\big)^2 = \mathrm{Var}_{y\sim p}[v_y]$, which gives Remark 1. In particular $C_p\mathbf{1} = 0$, so a uniform logit shift has zero score. Both Q_{recv} and the matched baseline

$$q_{p2}(r, \Delta u) = \tfrac{1}{2}\|JG_\ell(r)\Delta u\|_2^2 \quad (7)$$

use the same finite-difference estimate of $JG_\ell(r)\Delta u$; they differ only in the weighting by the receiver's next-token distribution.

B.3 First-order attribution of KL is zero

Let z_α be the logits after adding $\alpha\Delta u$, with $z_0 = z$. Differentiating (5) with $\Delta = z_\alpha - z$ gives $\frac{d}{d\alpha} D_{\mathrm{KL}}(p \,\|\, p_\alpha) = \langle p_\alpha - p, \dot{z}_\alpha\rangle$, which is zero at $\alpha = 0$. A first-order attribution score for KL is therefore identically zero at the native point. The AtP baselines instead use two non-stationary targets: the directional derivative of the native top-token log-probability and the directional derivative of the native top1–top2 logit margin. We take absolute values because the question is how strongly a receiver reacts along the intervention direction.

B.4 The three correlation estimands

Let $\mathcal{K}_d$ be the receivers of displacement d, including the original host i_d as an anchor, and let $X_{d,k}$ be a predictor.

1. **Pooled:** $\rho_{\mathrm{pool}} = \mathrm{Spearman}\big(\{X_{d,k}\}, \{K_{d,k}\}\big)$ over all rows, anchors included.
2. **Within-displacement (primary):** $\rho_d = \mathrm{Spearman}_{k\in\mathcal{K}_d\setminus\{i_d\}}(X_{d,k}, K_{d,k})$, reported as $\mathrm{median}_d\,\rho_d$ together with the fraction of displacements with $\rho_d > 0$. Displacements with fewer than four alternative receivers would be skipped; none are in the reported panels.
3. **Residual-rank (crossed matrices):** rank-transform X and K over all rows and standardize them, then remove displacement and receiver means by alternating demeaning until convergence. The statistic is the Pearson correlation of the two residual vectors.

For the ordinary host–donor matrices (Appendix D.1), the pair-specific statistic is computed in the same spirit, but on the original scale: $\log(1+\cdot)$-transformed values are two-way demeaned by host and donor, and the residuals are compared by Spearman correlation. We report each statistic with the method used to compute it in the frozen artifact; the two procedures are not numerically interchangeable.

B.5 Causal footprints and held-out receiver maps

For downstream layer m, the footprints $c_{d,i}^{(m)}$ and $c_{d,k}^{(m)}$ are differences between patched and native final-token hidden states (Equation (4)). Fidelity, gain, and the rewrite residual are defined in Section 4.2. For prediction, the 96 control-panel displacements are split once into 72 training and 24 held-out displacements, and the split is fixed before analysis. At each layer, a rank-16 PCA basis is fit to training footprints only, and ridge maps (regularization strength 1.0) are fit in this subspace. The baselines are the identity; a scalar gain per receiver; one global map using all receivers; the receiver-specific map; and a wrong-receiver map obtained by a fixed cyclic permutation of receiver identities. We report cosine prediction and normalized RMSE on held-out displacements. For uncertainty, receiver rows are averaged within each held-out displacement, so the 24 held-out displacements are the independent units for both the standard-error bands in Figure 3 and the bootstrap intervals in Section 5.6.

C Experimental and Reproducibility Details

C.1 Models, data, and intervention sites

We use Qwen3-8B and Mistral-7B-v0.3 with frozen weights. Contexts are sampled deterministically from the C4 validation split with SHA-256 thinning before model-specific tokenization. Each context contains 64 tokens, and all interventions and measurements use the final token. The four depth probes are Qwen layers 7, 14, 21, 26 and Mistral layers 6, 12, 19, 23, at relative depths of about 0.20, 0.40, 0.60, and 0.74. The primary depths are Qwen layer 14 and Mistral layer 12. Exact checkpoint revisions are recorded in the artifact.

C.2 Ordinary transplant protocol

The ordinary protocol uses 64 contexts and evaluates every directed non-self donor–host pair, which gives $64 \times 63 = 4032$ interventions at each tested depth. It supports host–donor residualization, the wrong-state diagnostic, and comparison with earlier activation-transplant settings. It does not isolate receiver causality by itself, because the displacement changes with the host. At the primary depth, two random subsets of this matrix are reused: 256 pairs for the dose curve (Figure 5) and 128 pairs for the teacher-forced check (Appendix D.10). A larger ordinary matrix covers all $256 \times 255 = 65{,}280$ directed pairs of the 256 contexts of bucket B1 in Mistral-7B at layer 12.

C.3 Crossed-receiver protocol

The crossed-receiver test uses three disjoint 256-context buckets, assigned by distinct SHA-256 residues modulo 11. Each stage samples 512 natural displacements from host–donor pairs within the bucket. In bucket B1, each displacement is crossed with 16 receivers (the original host as an anchor plus 15 alternatives), giving 8192 interventions per stage. B1 is repeated at all four depths in both models. Buckets B2 and B3 use 32 receivers (the anchor plus 31 alternatives) at the primary depth, giving 16,384 interventions each. In total, the twelve crossed stages contain 131,072 physical interventions. The anchor rows enter the pooled and residual statistics and the variance decomposition, but not the within-displacement statistic.

C.4 Control panel

The control panel is a separate deterministic 96×8 panel per model at the primary depth: 96 natural displacements, built from disjoint host and donor contexts, each crossed with the same eight receivers. Its contexts are selected by a different hash rule from the crossed buckets, and we do not assume that it is statistically independent of them. The panel stores layerwise final-token hidden-state differences, which the scalar-only crossed buckets do not. It is used for the alternative predictors, the equal-norm control, the causal footprints, and the held-out maps.

C.5 Alternative predictors and the equal-norm control

For every row of the control panel, we record Q_{recv}, q_{p2}, the two AtP scores (Appendix B.3), the receiver norm $\|r\|$, the relative displacement norm $\|\Delta u\|/\|r\|$, and $|\cos(\Delta u, r)|$. The AtP gradients are exact automatic-differentiation gradients of the respective targets with respect to the receiver state, with one backward pass through the suffix per receiver. For the physical equal-norm control, each natural displacement is rescaled to the median displacement norm of the panel before it is applied to its receivers, and exact KL is recomputed. Because a displacement's scale factor is shared by all of its receivers, the control asks whether the nonlinear receiver ordering changes once between-displacement magnitude differences are removed.

C.6 Implementation and compute

The large-model experiments use Hugging Face Transformers with FP16 weights, sharded with Accelerate across two NVIDIA T4 GPUs (16 GB each), with gradients disabled except for the AtP baselines, 64-token contexts, and pair batches of four. The twelve crossed-receiver stages and the 65,280-pair Mistral matrix have instrumented stage runtimes totalling 631.3 minutes (437.8 minutes crossed, 193.4 minutes large matrix). The eight ordinary 64-context depth targets total 134.9 minutes, as recorded in the artifact. The wall-clock time of the two control-panel runs was not instrumented, so we do not report it. The Pythia runs use FP32 and exact JVPs (Appendix D.11). The numerical audit of Appendix D.12.2 produces the natural states with the FP16 model on the same two T4 GPUs and then evaluates the downstream suffix in FP32, with TF32 disabled, split across both GPUs; its runtime was not recorded. Random seeds, context identifiers, and train/held-out splits are fixed before analysis. Frozen outputs, analysis scripts, and figure-generation code are available at `https://anonymous.4open.science/r/newinml-7402/`. All data figures in this paper are regenerated from these frozen outputs. Row-level outputs allow every crossed-receiver statistic and the Mistral control-panel intervals to be recomputed on CPU. For the Qwen3-8B control panel, the artifact stores the bootstrap and standard-error summaries but not per-displacement map scores.

D Additional Results

D.1 Ordinary transplants and the wrong-state diagnostic

At the primary depth, the ordinary-transplant Fisher–KL Spearman correlation over all 4032 pairs is 0.867 in Qwen3-8B and 0.911 in Mistral-7B. Removing additive host and donor effects leaves 0.563 and 0.677, with two-way bootstrap intervals $[0.484, 0.637]$ and $[0.607, 0.750]$. The wrong-state diagnostic keeps each transplant outcome fixed but evaluates the score of its displacement at unrelated states. On a random 256-pair subset, the correlation between the transplant KL and the score at the correct host is 0.875 in Qwen3-8B and 0.919 in Mistral-7B. When the score is evaluated at eight unrelated states per pair, it drops to -0.015 and 0.028 (2048 rows). The diagnostic shows that the score must be computed at the state that receives the update. It does not show that the receiving state causes the variation, because the displacement still changes with the host; the crossed design in the main text tests that directly.

For the larger Mistral matrix of 65,280 ordinary transplants, the raw Fisher–KL correlation is 0.911, the host–donor residual correlation is 0.660, and the corresponding q_{p2} residual correlation is 0.415. A joint context-label permutation test, which applies the same permutation to both matrix axes, gives $p = 0.0005$, the smallest value attainable with 2000 permutations.

D.2 Rank-KL variance decomposition

To compare the correlation summaries of Section 3.2, we fit descriptive fixed-effect models to the rank-transformed exact KL in each crossed matrix (Table 4). Receiver identity alone explains most of the rank variance, which is why the within-displacement correlation can be high even though it is not purely pair-specific. Adding displacement identity increases the in-sample additive R^2 further, but 18–25% remains. This remainder is the variation targeted by the residual-rank correlation. Because there is one observed outcome per displacement–receiver cell, we call it non-additive residual variation, not a separately identified interaction.

Table 4: **Descriptive decomposition of rank-transformed exact KL in the six crossed matrices.** "Receiver" and "Displacement" are one-factor fixed-effect R^2 values; "Additive" uses both factors; "Non-additive" is $1 - R^2$ of the additive model. N includes the anchor rows.

Model	Bucket	N	Receiver R^2	Displacement R^2	Additive R^2	Non-additive
Qwen3-8B	B1	8192	0.707	0.123	0.789	0.211
Qwen3-8B	B2	16384	0.733	0.063	0.779	0.221
Qwen3-8B	B3	16384	0.704	0.071	0.754	0.246
Mistral-7B	B1	8192	0.773	0.076	0.804	0.196
Mistral-7B	B2	16384	0.795	0.041	0.815	0.185
Mistral-7B	B3	16384	0.798	0.044	0.820	0.180

D.3 Reconciling the Fisher–KL correlations reported in this paper

The paper reports several Fisher–KL correlations. They differ because they use different estimands, designs, samples, and numbers of receivers per displacement, not because of inconsistent reruns. Table 5 lists them side by side. Three patterns explain the differences. First, within-displacement medians over 8 receivers (control panel) are coarser and lower than those over 15 or 31 receivers (crossed buckets). Second, residual statistics remove the dominant receiver main effect and are therefore lower than within-displacement or pooled ones (Table 4). Third, random subsets of the ordinary matrix (256 dose pairs, 256 wrong-state pairs, 128 teacher-forced pairs) differ from the full 4032-pair value by ordinary sampling variation; at full dose, the dose-subset KL is the same quantity as the ordinary transplant KL.

Table 5: **Every Fisher–KL correlation in the paper, with its design, sample, and estimand.** Ranges are over buckets or depths. All values come from the frozen artifact.

Design and sample	Receivers per Δu	Estimand	Qwen3-8B	Mistral-7B
Crossed B1–B3, primary depth	15 / 31 alternatives	within (primary)	0.858–0.872	0.911–0.928
Crossed B1–B3, primary depth	anchor + 15 / 31	pooled	0.855–0.871	0.921–0.932
Crossed B1–B3, primary depth	anchor + 15 / 31	residual-rank	0.504–0.520	0.658–0.675
Crossed B1, four depths	15 alternatives	within	0.857–0.954	0.854–0.957
Control panel (96×8)	8	within	0.738	0.905
Control panel (96×8)	8	pooled	0.722	0.876
Control panel (96×8)	8	residual-rank	0.456	0.612
Ordinary, all 4032 pairs	—	pooled	0.867	0.911
Ordinary, all 4032 pairs	—	host–donor resid.	0.563	0.677
Ordinary, 256 wrong-state pairs	—	pooled, correct state	0.875	0.919
Ordinary, 256 dose pairs, $\lambda = 1$	—	pooled	0.831	0.907
Ordinary, 128 pairs, $h = 1$	—	pooled	0.857	0.930
Large ordinary, 65,280 pairs	—	pooled / resid.	—	0.911 / 0.660

D.4 Depth sweep

Table 6 gives the B1 crossed-receiver result at every probed depth. The receiver-local score ranks receivers at all four depths in both models, and the within-displacement correlation is positive for at least 99.4% of displacements at every depth.

Table 6: **Crossed-receiver result across depth (bucket B1; 512 displacements, 15 alternative receivers each).** Median within-displacement Spearman correlation between Q_{recv} and exact KL, and the fraction of displacements with a positive correlation.

Model	Layer	Relative depth	Median within $\rho \uparrow$	Positive fraction
Qwen3-8B	7	0.20	0.857	1.000
Qwen3-8B	14	0.40	0.871	0.994
Qwen3-8B	21	0.60	0.936	1.000
Qwen3-8B	26	0.74	0.954	1.000
Mistral-7B	6	0.19	0.854	1.000
Mistral-7B	12	0.39	0.911	1.000
Mistral-7B	19	0.61	0.929	0.994
Mistral-7B	23	0.74	0.957	1.000

D.5 Spread of the effect of a fixed displacement

For each crossed displacement, we compute $\log_{10}(\max_k K_{d,k} / \min_k K_{d,k})$ over its alternative receivers (Table 7). The spread grows with the number of receivers, as any range statistic does, so we compare buckets with the same receiver count.

Table 7: **How much the exact effect of one frozen displacement varies across its receivers.** Median over 512 displacements of $\log_{10}$ of the ratio between the largest and smallest exact KL among the alternative receivers, and the fraction of displacements whose ratio is at least 100.

Model	Bucket	Alternative receivers	Median $\log_{10}$ max/min	Fraction with ratio ≥ 100
Qwen3-8B	B1	15	3.46	0.916
Qwen3-8B	B2	31	4.77	0.994
Qwen3-8B	B3	31	4.36	0.990
Mistral-7B	B1	15	2.90	0.865
Mistral-7B	B2	31	3.31	0.994
Mistral-7B	B3	31	3.65	1.000

D.6 Full alternative-predictor panel

The compact main-text Table 2 reports the primary within-displacement statistic. Table 8 adds the pooled correlation, the fraction of displacements with a positive within-displacement correlation, and the residual-rank correlation. The residual is undefined for the receiver norm, because a predictor that depends only on the receiver is removed entirely by the receiver main effect.

Table 8: **Alternative predictors on the** 96×8 **control panel** (768 rows per model). The magnitudes differ across columns because the estimands differ (Section 3.2).

Model	Predictor	Pooled ρ	Within median ρ	Positive fraction	Residual ρ
Qwen3-8B	Fisher Q_{recv}	0.722	0.738	0.979	0.456
Qwen3-8B	AtP-top	0.512	0.500	0.906	0.283
Qwen3-8B	AtP-margin	0.434	0.429	0.875	0.222
Qwen3-8B	q_{p2}	0.261	0.226	0.708	0.237
Qwen3-8B	$\|\Delta u\|/\|r\|$	0.350	0.071	0.615	0.108
Qwen3-8B	$\|r\|$	-0.081	-0.071	0.354	—
Qwen3-8B	$\|\cos(\Delta u, r)\|$	-0.056	-0.012	0.479	0.047
Mistral-7B	Fisher Q_{recv}	0.876	0.905	1.000	0.612
Mistral-7B	AtP-top	0.736	0.750	1.000	0.413
Mistral-7B	AtP-margin	0.156	0.179	0.656	0.407
Mistral-7B	q_{p2}	0.089	0.060	0.552	0.197
Mistral-7B	$\|\Delta u\|/\|r\|$	0.118	0.000	0.458	-0.036
Mistral-7B	$\|r\|$	-0.006	0.000	0.479	—
Mistral-7B	$\|\cos(\Delta u, r)\|$	0.030	0.048	0.531	0.003

D.7 Layerwise causal footprints

Figure 4 shows how the footprints of one displacement at the original host and at a receiver separate across downstream layers. Fidelity and gain both start at one at the intervention site and decay toward zero by the final layer. The receiver's footprint is therefore neither a rescaled copy of the host's nor simply attenuated along the host's direction; most of it lies in a different direction.

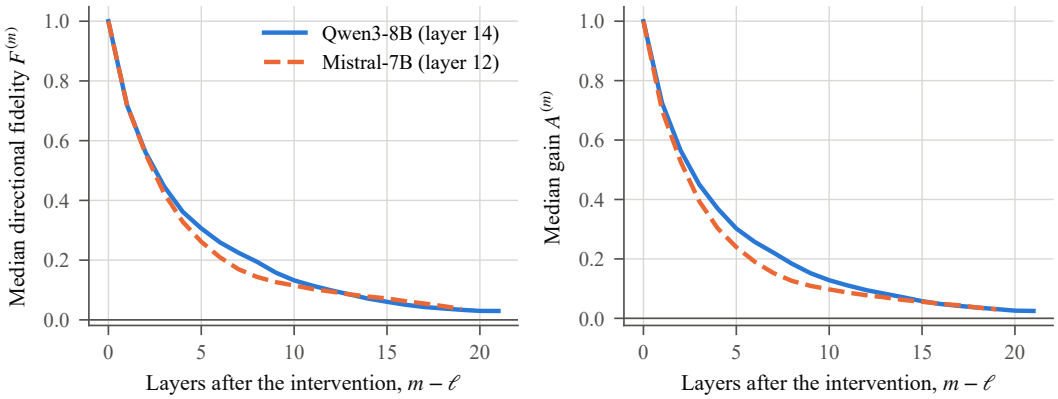


Figure 4: **The downstream direction of a fixed displacement diverges across receivers.** Median over the 768 displacement–receiver pairs of the 96×8 control panel of the directional fidelity $F^{(m)}$ (left) and the gain $A^{(m)}$ (right) between the original host's and the receiver's causal footprints, by number of layers after the intervention. At the final layer, median fidelity is 0.030 in Qwen3-8B and 0.037 in Mistral-7B.

D.8 Receiver-specific transformation at the final layer

Table 9 reports the final-layer held-out prediction scores underlying Figure 3. The receiver-specific map improves directional prediction much more than exact vector reconstruction. This supports a modest interpretation: receiver identity contributes a reproducible directional component to the downstream transformation, while substantial update-specific variation remains unexplained.

Table 9: **Held-out final-layer prediction** (Qwen3-8B layer 35, Mistral-7B layer 31; 24 held-out displacements $\times$ 8 receivers). Cosine is averaged over receiver–displacement pairs; lower normalized RMSE is better. Bold marks the best predictor per model.

Model	Predictor	Mean cosine ↑	Mean NRMSE ↓
Qwen3-8B	Identity	0.029	1.823
Qwen3-8B	Gain only	0.021	1.000
Qwen3-8B	Global map	0.086	1.005
Qwen3-8B	Receiver-specific map	**0.316**	**0.945**
Qwen3-8B	Wrong-receiver map	−0.023	1.122
Mistral-7B	Identity	0.037	1.859
Mistral-7B	Gain only	0.037	1.000
Mistral-7B	Global map	0.109	0.999
Mistral-7B	Receiver-specific map	**0.241**	**0.986**
Mistral-7B	Wrong-receiver map	0.028	1.080

D.9 Intervention dose

On a random 256-pair subset of the ordinary matrix at the primary depth, we apply $u_i + \lambda \Delta u_{ij}$ for $\lambda \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$ and correlate Q_{recv} with the exact KL at each dose (Figure 5). The correlation is highest at small doses, where the local expansion of Appendix B.1 is most accurate, and it remains high at the full transplant.

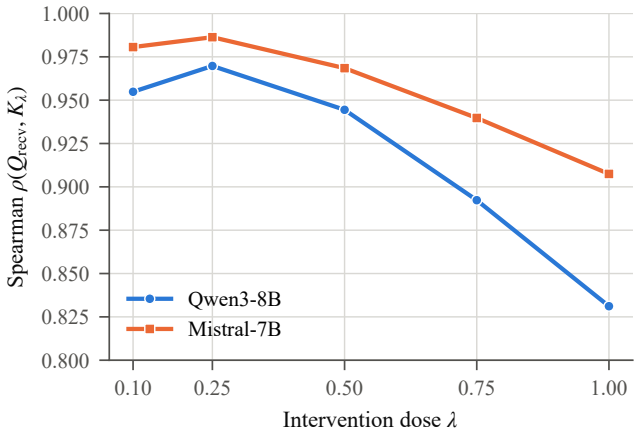


Figure 5: **The local score remains informative as the intervention reaches full dose.** Spearman correlation between Q_{recv} and exact KL at dose λ on 256 ordinary transplant pairs per model at the primary depth. Values at $\lambda = 0.10, 0.25, 0.50, 0.75, 1.00$: $0.955, 0.970, 0.944, 0.892, 0.831$ in Qwen3-8B and $0.981, 0.986, 0.968, 0.940, 0.907$ in Mistral-7B.

D.10 Teacher-forced persistence check

The main experiments measure the next-token distribution. To check whether the association extends beyond the immediate next token, we take a random 128-pair subset of the ordinary matrix at the primary depth and run each patched forward pass on the true C4 continuation. The MLP output is replaced at the final context token only, and later positions see the change only through the model's own computation. We sum the per-position KL over the next $h \in \{1, 4, 8, 16, 32\}$ positions and correlate Q_{recv} with this cumulative disturbance (Table 10 and Figure 6); $h = 1$ is the ordinary next-token KL. Receiver-local sensitivity remains associated with cumulative disturbance beyond the immediate next token, although the association weakens with the horizon, especially in Qwen3-8B. Teacher forcing fixes the continuation, so this check does not establish persistence under unconstrained autoregressive generation.

Table 10: **Teacher-forced persistence.** Spearman correlation between Q_{recv} and KL summed over the next h teacher-forced positions (128 ordinary transplant pairs per model, primary depth).

Model	$h = 1$	$h = 4$	$h = 8$	$h = 16$	$h = 32$
Qwen3-8B	0.857	0.771	0.704	0.614	0.565
Mistral-7B	0.930	0.896	0.839	0.819	0.812

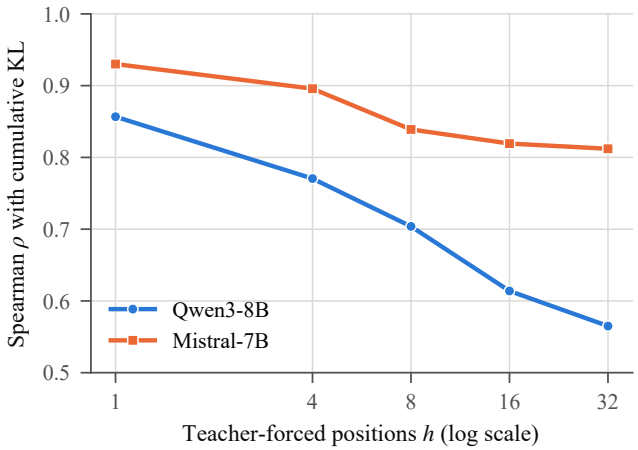


Figure 6: **Association with cumulative disturbance under teacher forcing.** The values of Table 10 plotted against the horizon h (log scale). The association persists beyond the next token and weakens with h.

D.11 Independent-pretraining replication (Pythia-70M, ordinary transplants)

This replication uses ordinary transplants, not the crossed-receiver design. It tests whether the association between local geometry and transplant KL replicates across independently pretrained models. We use the final checkpoints (step 143,000) of Pythia-70M and five further pretraining seeds [Biderman et al., 2023, van der Wal et al., 2025], with MLP layer 2 and FP32 arithmetic. For each run, 256 hosts are drawn from a held-out pool of C4 contexts (corpus B). Each host receives one donor chosen deterministically from its five nearest non-self neighbours among the MLP-input states, after a 32-dimensional PCA. The downstream Fisher score Q_{recv} is computed with exact automatic-differentiation JVPs. The reported statistic is a rank-residualized partial Spearman correlation between $\log(1 + Q_{\mathrm{recv}})$ and transplant KL. Both variables are rank-transformed, and the ranks of four controls are regressed out by least squares: the log update distance $\log(1 + \|\Delta u\|)$, the log state distance $\log(1 + \|\Delta m\|)$ between donor and host MLP inputs, the normalized nonlinear residual $\|\Delta u - J_{F_\ell}(m_i)\Delta m\|/\|\Delta u\|$, and the tangent alignment $\cos(J_{F_\ell}(m_i)\Delta m, \Delta u)$, where J_{F_ℓ} is the exact MLP Jacobian at the host. Table 11 reports the six runs. All six are positive, which shows that the local-geometry association under ordinary transplants is not specific to one pretraining run.

Table 11: **Partial Spearman correlation between** $\log(1 + Q_{\mathrm{recv}})$ **and transplant KL on six independently pretrained Pythia-70M runs** (ordinary transplants, layer 2, 256 hosts per run, controls as described in Appendix D.11). The interval is a bootstrap over the six runs.

Run	Hosts	Partial ρ
base	256	0.775
seed1	256	0.824
seed2	256	0.723
seed3	256	0.828
seed4	256	0.762
seed5	256	0.825
Median	—	0.800
95% bootstrap interval	—	[0.742, 0.827]

D.12 Numerical audit of the finite-difference score

The crossed experiments estimate the directional response $JG_\ell(r)\Delta u$ by centered finite differences at $\epsilon = 0.075$ on FP16 model evaluations. We audit this estimate on one frozen 16-pair panel per model: the first 16 control-panel displacements (Appendix C.4), each applied to receiver 0. The 16-pair audit varies displacement direction at a fixed receiver and therefore isolates numerical closure of the

local directional derivative; it does not reproduce the primary within-displacement receiver-ranking statistic, which rests on the physical full-dose crossed experiments. We ask two separate questions.

D.12.1 End-to-end FP16 step-size stability

On FP16 model evaluations, we compare the response at ϵ with the response at half step $\epsilon/2 = 0.0375$ and with centered-difference Richardson extrapolation. The median cosine between the responses at ϵ and $\epsilon/2$ is 0.989 for Qwen3-8B and 0.940 for Mistral-7B, and between the ϵ response and the Richardson estimate 0.981 and 0.899. The median relative difference of the Fisher score from the Richardson estimate is 0.333 and 0.472. These end-to-end comparisons combine finite-difference step effects, FP16 finite-precision effects, and the corresponding changes in native logits and probabilities, so they should not be read as truncation error alone. In the ordinary-matrix step-size sweep, rank stability between step sizes is above 0.95 for every comparison involving the chosen $\epsilon = 0.075$; the minimum over all compared pairs of step sizes is 0.949 in Qwen3-8B (for $\epsilon = 0.04$ versus 0.05) and 0.961 in Mistral-7B.

D.12.2 Exact FP32 downstream-suffix closure

To separate truncation error from finite precision, we recompute the same 16 cases with an exact FP32 downstream-suffix JVP from the same cached natural receiver states. The natural states up to and including layer ℓ (the host and donor MLP outputs and the receiver's layer-ℓ output) are produced in FP16, as in the main experiments. The downstream suffix, from layer $\ell + 1$ through the final norm and the language-model head, is then evaluated in FP32 with TF32 disabled. On this suffix, we compute the exact JVP by automatic differentiation and the centered finite differences at ϵ and $\epsilon/2$, with their Richardson combination. The upstream computation is not repeated in FP32, so this is not a full-model FP32 JVP. The host, donor, and receiver context identifiers match the frozen control panel for all 16 cases in both models.

Table 12: **Numerical audit on 16 frozen cases per model** (control-panel displacements 0–15 at receiver 0; Qwen3-8B layer 14, Mistral-7B layer 12). The upper block compares FP32 finite differences with exact FP32 suffix JVPs. The lower block compares the FP16 finite-difference scores for these same 16 control-panel cases, computed with the same estimator used in the crossed experiments, with the exact FP32 suffix-JVP scores. "Relative Q error" is $|Q - Q_{\mathrm{JVP32}}|/Q_{\mathrm{JVP32}}$.

Comparison	Statistic	Qwen3-8B	Mistral-7B
FD32 ($\epsilon = 0.075$) vs. JVP32	median directional cosine	1.0000	1.0000
	Spearman $\rho(Q_{\mathrm{FD32}}, Q_{\mathrm{JVP32}})$	1.000	1.000
	median relative Q error	0.21%	0.20%
	maximum relative Q error	0.82%	0.37%
FD32 ($\epsilon/2$) vs. JVP32	median relative Q error	0.12%	0.07%
FD32 Richardson vs. JVP32	median relative Q error	0.18%	0.13%
FP16 FD ($\epsilon = 0.075$) vs. JVP32	Spearman $\rho(Q_{\mathrm{FP16}}, Q_{\mathrm{JVP32}})$	0.785	0.544
	median relative Q difference	21.6%	26.7%

Table 12 reports the result. In FP32, the centered finite difference at $\epsilon = 0.075$ reproduces the exact JVP almost perfectly: every directional cosine is at least 0.99998, the Fisher scores are identically ranked, and the median relative score error is about 0.2% in both models, with the half step and Richardson combination equally close. The step $\epsilon = 0.075$ itself therefore introduces negligible centered-difference truncation error on this panel. The FP16 finite-difference scores for the same 16 cases agree less closely with the exact FP32 scores (Spearman 0.785 and 0.544; median relative difference 21.6% and 26.7%), and Richardson extrapolation of the FP16 differences moves further from them (Spearman 0.697 and 0.474), consistent with amplification of finite-precision noise. On this audit panel, this pattern is consistent with finite-precision effects dominating the observed discrepancy in Appendix D.12.1 rather than centered-difference truncation error. Accordingly, we treat the large-scale FP16 Fisher values as ranking diagnostics rather than as numerically exact FP32 quadratic forms; the audit closes truncation error on the tested directions but does not make the FP16 scores exact. The causal-footprint and held-out receiver-map results in Sections 5.5 and 5.6 do not depend on the finite-difference Fisher estimator.

E Scope and Interpretation

The crossed design supports a causal statement about the receiving state, because the intervention vector is physically held fixed within each displacement. It does not establish that an MLP update represents a context-independent semantic function, and we make no such claim. Likewise, the low-rank maps should not be read as complete or deterministic receiver operators. Their held-out cosines are meaningful but far from one, and their normalized RMSE gains are modest. The result is better read as evidence that a reproducible receiver-specific component coexists with substantial update-specific variation.

The next-token endpoint is deliberately narrow. It gives a precise, full-vocabulary behavioral measure and makes exact physical controls possible, but it does not characterize free-running generation. The paper is diagnostic and proposes no steering or editing method. Natural extensions include longer-horizon behavior, other component types such as attention outputs, mechanistic decomposition of the receiver dependence, and intervention methods that aim to preserve a known causal effect across receiver states.

References

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, Usvsn Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. Pythia: A suite for analyzing large language models across training and scaling. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2397–2430. PMLR, 23–29 Jul 2023.

Arthur Conmy, Augustine Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems*, volume 36, pages 16318–16352, 2023. doi: 10.52202/075280-0719. URL https://papers.neurips.cc/paper_files/paper/2023/hash/34e1dbe95d34d7ebaf99b9bcaeb5b2be-Abstract-Conference.html.

Weilong Dong, Xinwei Wu, Renren Jin, Shaoyang Xu, and Deyi Xiong. CONTRANS: Weak-to-strong alignment engineering via concept transplantation. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 4130–4148, Abu Dhabi, UAE, 2025. Association for Computational Linguistics. URL https://aclanthology.org/2025.coling-main.279/.

Gabriel Garcia. No free swap: Protocol-dependent layer redundancy in transformers. *arXiv preprint arXiv:2605.16234*, 2026.

Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 9574–9586. Curran Associates, Inc., 2021.

Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan, Jing Huang, Aryaman Arora, Zhengxuan Wu, Noah Goodman, Christopher Potts, and Thomas Icard. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64, 2025. URL https://jmlr.org/papers/v26/23-0058.html.

Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.446.

Mor Geva, Avi Caciularu, Kevin Wang, and Yoav Goldberg. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 30–45, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.3.

Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 15466–15490. PMLR, 2024. URL https://proceedings.mlr.press/v235/ghandeharioun24a.html.

Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching. *arXiv preprint arXiv:2304.05969*, 2023. URL https://arxiv.org/abs/2304.05969.

Satchel Grant, Simon Jerome Han, Alexa R. Tartaglini, and Christopher Potts. Addressing divergent representations from causal interventions on neural networks. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=cZrTMqYVL6.

János Kramár, Tom Lieberum, Rohin Shah, and Neel Nanda. AtP*: An efficient and scalable method for localizing LLM behaviour to components. *arXiv preprint arXiv:2403.00745*, 2024. URL https://arxiv.org/abs/2403.00745.

Jiaqian Li, Yanshu Li, and Kuan-Hao Huang. Steering vector fields for context-aware inference-time control in large language models. *arXiv preprint arXiv:2602.01654*, 2026.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.52202/075280-1797.

SiYuan Ma, Yiqin Luo, Zhangji, Canran Xiao, Albert Gao, Wei-Hsing Huang, Wei Wang, Qiwei Wu, Xinran Li, Jinfeng Wei, and Qixin Zhang. Hidden APIs in language models: Discovering reusable causal interfaces from forked futures. *arXiv preprint arXiv:2607.27617*, 2026. URL https://arxiv.org/abs/2607.27617.

James Martens. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21(146):1–76, 2020.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. doi: 10.52202/068431-1262.

David N. Olivieri and Antonio F. Pérez Rodríguez. Transformer field theory: A response-theoretic approach to mechanistic interpretability. *arXiv preprint arXiv:2605.25225*, 2026.

Gustaw Opiełka, Hannes Rosenbusch, and Claire E. Stevenson. Causality $\neq$ invariance: Function and concept vectors in LLMs. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/86b3697c4eb7792c951831636bfdacd5-Abstract-Conference.html.

Koyena Pal, Jiuding Sun, Andrew Yuan, Byron C. Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, pages 548–560, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.conll-1.37. URL https://aclanthology.org/2023.conll-1.37/.

Kiho Park, Todd Nief, Yo Joong Choe, and Victor Veitch. The information geometry of softmax: Probing and steering. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL https://arxiv.org/abs/2602.15293.

Philip Quirke. Necessary, decodable and reversible, yet not transferable: A stress test for attention-head role claims. *arXiv preprint arXiv:2606.08292*, 2026.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.828.

Shashwat Singh, Shauli Ravfogel, Jonathan Herzig, Roee Aharoni, Ryan Cotterell, and Ponnurangam Kumaraguru. Representation surgery: Theory and practice of affine steering. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 45663–45680. PMLR, 2024. URL https://proceedings.mlr.press/v235/singh24d.html.

Julian Skifstad, Xinyue Annie Yang, and Glen Chou. Local linearity of LLMs enables activation steering via model-based linear optimal control. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL https://arxiv.org/abs/2604.19018.

Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. Function vectors in large language models. In *The Twelfth International Conference on Learning Representations*, 2024.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023. URL https://arxiv.org/abs/2308.10248.

Sankaran Vaidyanathan, David Arbour, Aaron Mueller, Scott Niekum, and David Jensen. The curse of multiple mediators: Hidden interaction effects in activation patching. *arXiv preprint arXiv:2606.27510*, 2026.

Oskar van der Wal, Pietro Lesci, Max Müller-Eberstein, Naomi Saphra, Hailey Schoelkopf, Willem Zuidema, and Stella Biderman. PolyPythias: Stability and outliers across fifty language model pre-training runs. In *The Thirteenth International Conference on Learning Representations*, 2025.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart M. Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/hash/92650b2e92217715fe312e6fa7b90d82-Abstract.html.

Sihan Wang and Jiayi Zhao. FishBack: Pullback fisher geometry for optimal activation steering in transformers. *arXiv preprint arXiv:2605.17231*, 2026. URL https://arxiv.org/abs/2605.17231.

Zijian Wang, Britney Whyte, and Chang Xu. Locating and extracting relational concepts in large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 4818–4832, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.287. URL https://aclanthology.org/2024.findings-acl.287/.

Zhengxuan Wu, Atticus Geiger, Thomas Icard, Christopher Potts, and Noah Goodman. Interpretability at scale: Identifying causal mechanisms in Alpaca. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.52202/075280-3418. URL https://papers.nips.cc/paper_files/paper/2023/hash/f6a8b109d4d4fd64c75e94aaf85d9697-Abstract-Conference.html.

Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL https://arxiv.org/abs/2310.01405.